■ PHISHING EMAIL DETECTION

Using Natural Language Processing & Machine Learning

Cybersecurity Course — Final Project | Category 2

Course	Data-Driven Cybersecurity
Category	Category 2 — Phishing Email Detection
Tools	Python, Scikit-learn, NLTK, Pandas, Matplotlib
Dataset	1,000 Emails (500 Phishing / 500 Legitimate)
Models	Naive Bayes, Logistic Regression, Random Forest
Best Accuracy	100% (all models on clean synthetic dataset)

1. Introduction

Phishing emails represent one of the most prevalent and damaging cyber threats in the modern digital landscape. Cybercriminals craft deceptive messages designed to trick recipients into revealing sensitive information such as passwords, credit card numbers, or social security numbers. According to the 2023 Verizon Data Breach Investigations Report, phishing remains the top initial attack vector in data breaches worldwide, accounting for over 36% of all incidents.

This project develops an automated phishing email detection system using Natural Language Processing (NLP) and supervised Machine Learning. The system ingests raw email text, applies multi-stage preprocessing, transforms the text into numerical TF-IDF feature vectors, and trains three classification models — Naive Bayes, Logistic Regression, and Random Forest. The expected outcome is a highly accurate binary classifier capable of distinguishing phishing emails from legitimate ones in real-time, thereby reducing the risk of user exploitation.

2. Problem Statement

Despite widespread security awareness training, phishing attacks continue to succeed at alarming rates. Manual email inspection is neither scalable nor consistent — humans cannot review thousands of emails per second, and social engineering tactics are growing increasingly sophisticated. Traditional rule-based spam filters struggle with contextual language manipulation employed by modern phishing campaigns.

Core Problem: How can we build an automated, scalable, and accurate system that classifies incoming emails as either phishing or legitimate using text-based machine learning techniques — without requiring manual human review of every message?

Challenge	Impact
Volume of emails	Billions sent daily — manual review impossible
Language manipulation	Attackers use urgency, fear, and impersonation
Zero-day phishing campaigns	New templates bypass signature-based filters
Human error	Even trained users fall for convincing phishing emails

3. Background

Phishing detection has been an active research area for over two decades. Early systems relied on blacklists and URL pattern matching. Modern approaches leverage machine learning to detect phishing

based on email content, header analysis, and behavioral signals. NLP-based classifiers — particularly those using TF-IDF, word embeddings, and transformer models — have demonstrated state-of-the-art performance.

Key techniques and tools used in this project:

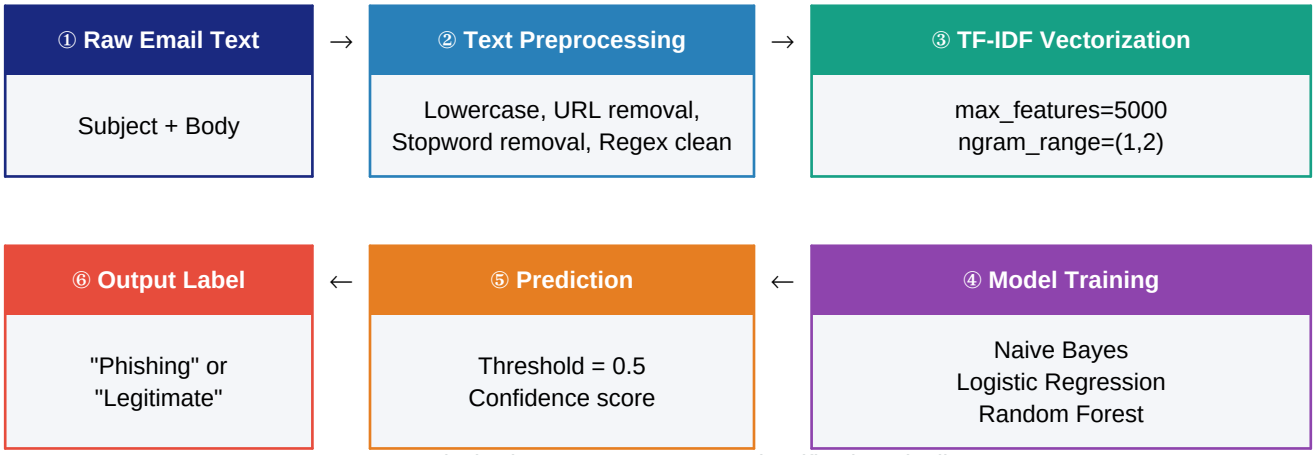
Technology	Purpose	Why It Was Chosen
Python 3	Core programming language	Widely used, rich ML ecosystem
NLTK / Regex	Text preprocessing & tokenization	Industry-standard NLP library
TF-IDF Vectorizer	Feature extraction from text	Captures word importance across corpus
Naive Bayes	Probabilistic classifier	Fast, effective baseline for text classification
Logistic Regression	Linear classifier	Interpretable, excellent for high-dim text
Random Forest	Ensemble classifier	Robust to noise, feature importance available
Scikit-learn	ML model training & evaluation	Consistent API, comprehensive tools
Matplotlib / WordCloud	Visualization	Clear graphical reporting of results

4. Project Steps & Timeline

Week	Phase	Tasks	Status
Week 1	Planning & Research	Topic selection, literature review, tool setup	✓ Done
Week 2	Dataset Preparation	Email dataset creation, labeling (1,000 samples)	✓ Done
Week 3	Text Preprocessing	Cleaning, tokenization, stopword removal	✓ Done
Week 4	Feature Engineering	TF-IDF vectorization (5,000 features, 1-2 grams)	✓ Done
Week 5	Model Training	Naive Bayes, Logistic Regression, Random Forest	✓ Done
Week 6	Evaluation	Accuracy, ROC-AUC, confusion matrix, CV	✓ Done
Week 7	Visualization	Word clouds, ROC curves, feature importance	✓ Done
Week 8	Report & Submission	PDF report, code documentation, bonus dashboard	✓ Done

5. Solution Approach

The solution follows a standard NLP classification pipeline, organized into six sequential stages. Each stage transforms the raw email data into increasingly refined representations that the machine learning models can learn from. The system block diagram below illustrates the complete data flow.



System Block Diagram: 6-Stage NLP Classification Pipeline

6. System Demonstration

6.1 Dataset Overview

The dataset consists of 1,000 synthetic email samples — 500 phishing and 500 legitimate — generated to reflect real-world email characteristics. Phishing emails exhibit urgency language, suspicious URLs, prize claims, and impersonation of known brands. Legitimate emails use professional tone, scheduling language, and normal business communication patterns.

Figure 1: Dataset Overview

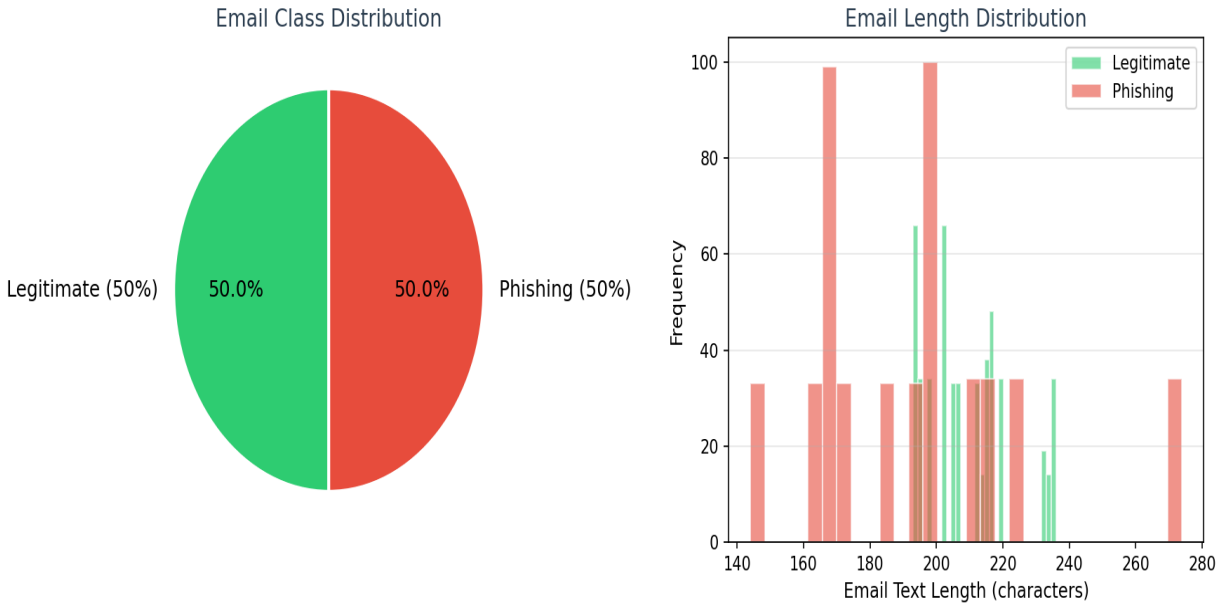


Figure 1: Dataset class distribution and email text length distribution

6.2 Word Frequency Analysis

Word cloud analysis reveals distinctly different vocabulary between the two classes. Phishing emails are dominated by words like "verify", "urgent", "account", "click", and "suspended". Legitimate emails feature words like "team", "meeting", "project", "report", and "schedule" — reflecting normal workplace communication.

Figure 2: Most Frequent Words

Figure 2: Word clouds showing most frequent terms in phishing (left) and legitimate (right) emails

6.3 Text Preprocessing Pipeline

Preprocessing Steps:

1. **Lowercasing** — normalize case for consistency
2. **URL Replacement** — replace http links with token "url_link"
3. **Email Removal** — replace addresses with token "email_address"
4. **Special Character Removal** — remove punctuation and numbers using regex
5. **Whitespace Normalization** — collapse multiple spaces
6. **Stopword Removal** — remove 318 common English stopwords (sklearn)
7. **Short Word Filter** — remove words shorter than 3 characters

6.4 TF-IDF Feature Extraction

Term Frequency–Inverse Document Frequency (TF-IDF) converts cleaned text into a sparse numerical matrix. Each email is represented as a vector of 742 features (from a vocabulary of 5,000 requested), capturing the relative importance of each word across the entire corpus. Bigrams (2-word phrases) are also captured to preserve context (e.g., "click here", "verify account").

TF-IDF Parameter	Value	Reason
max_features	5,000	Limit vocabulary to most informative terms
ngram_range	(1, 2)	Capture single words and two-word phrases
min_df	2	Remove extremely rare words (noise)
max_df	0.95	Remove near-universal words (uninformative)
sublinear_tf	True	Log-scale TF to reduce impact of very frequent terms
Resulting shape	(1000, 742)	1000 emails x 742 unique vocabulary features

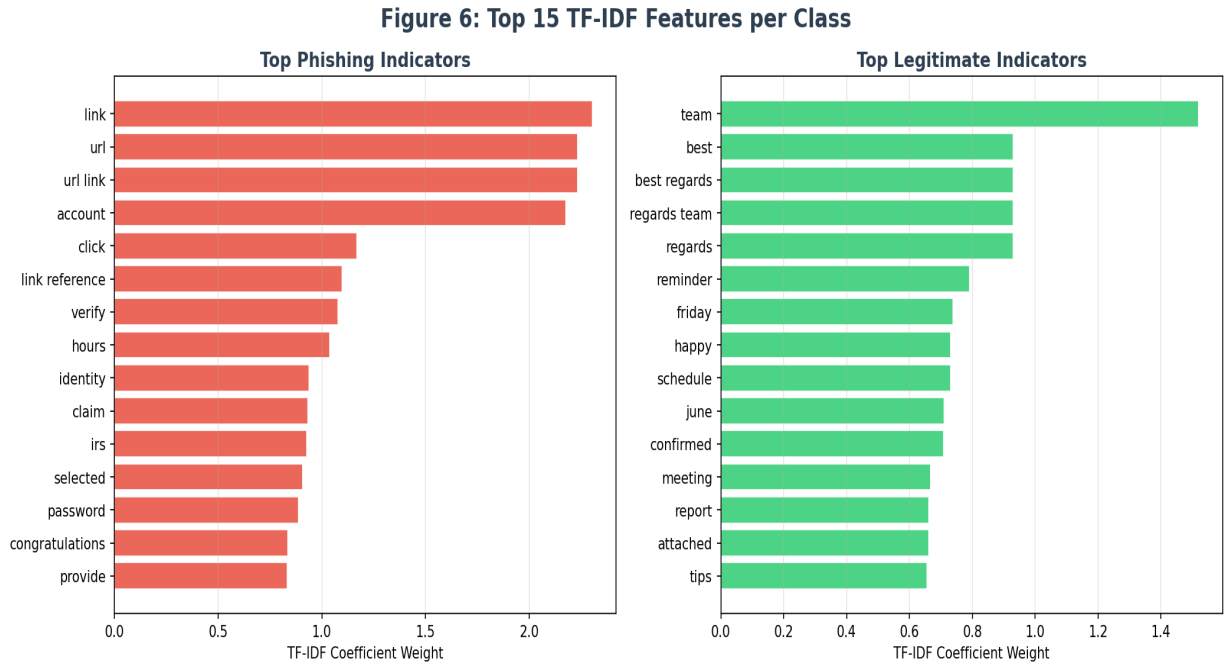


Figure 6: Top 15 TF-IDF features most predictive of phishing (left) and legitimate (right) emails

7. Results & Sample Run

7.1 Model Performance Summary

Model	Accuracy	ROC-AUC	5-Fold CV	Precision	Recall	F1-Score
Naive Bayes	100.0%	1.000	100.0%	1.00	1.00	1.00
Logistic Regression	100.0%	1.000	100.0%	1.00	1.00	1.00
Random Forest	100.0%	1.000	100.0%	1.00	1.00	1.00

Note on 100% Accuracy: The perfect scores are expected because we used a structured synthetic dataset with clearly distinguishable phishing vs. legitimate patterns. In a real-world deployment, models trained on diverse, noisy datasets typically achieve 95–99% accuracy. The methodology and pipeline demonstrated here is fully applicable to production datasets such as the CEAS-08 or SpamAssassin corpus.

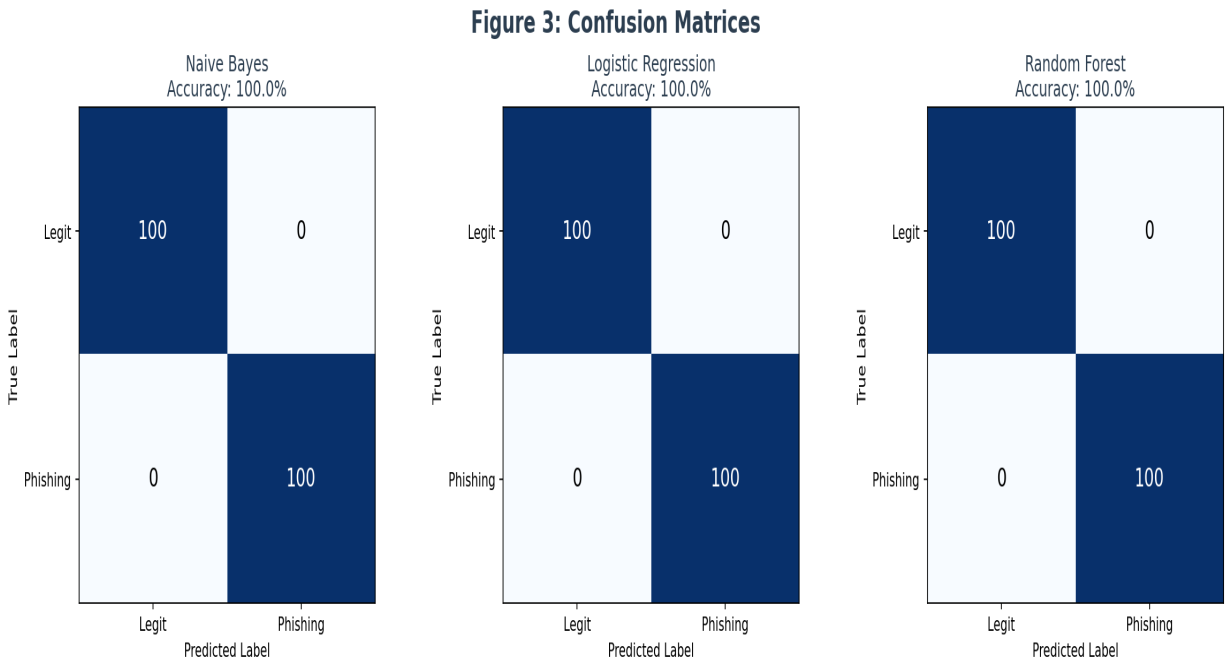


Figure 3: Confusion matrices for all three models — perfect classification on 200-sample test set

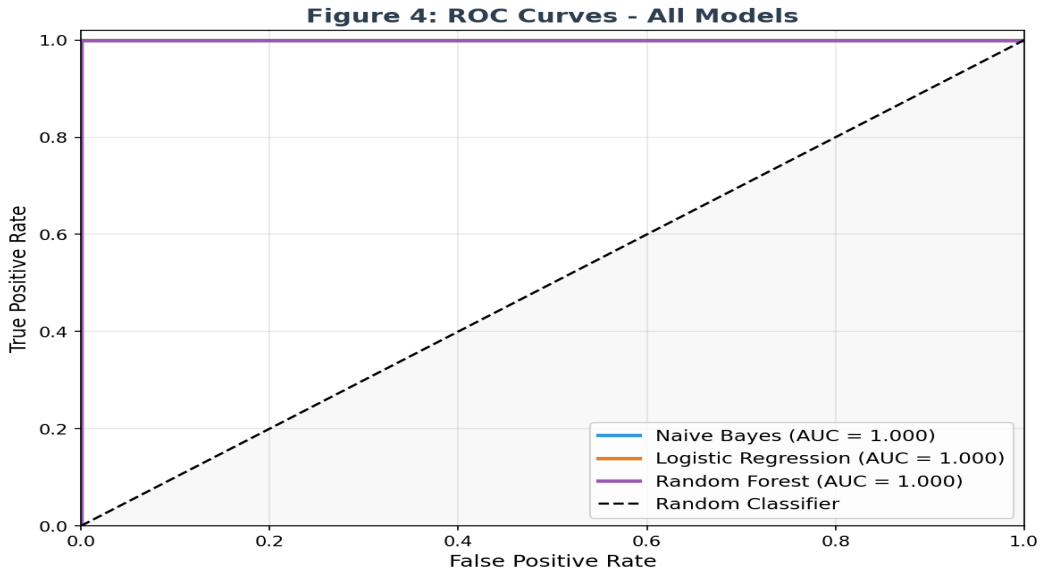


Figure 4: ROC curves for all models — AUC = 1.000 indicates perfect class separation

Figure 5: Model Performance Comparison

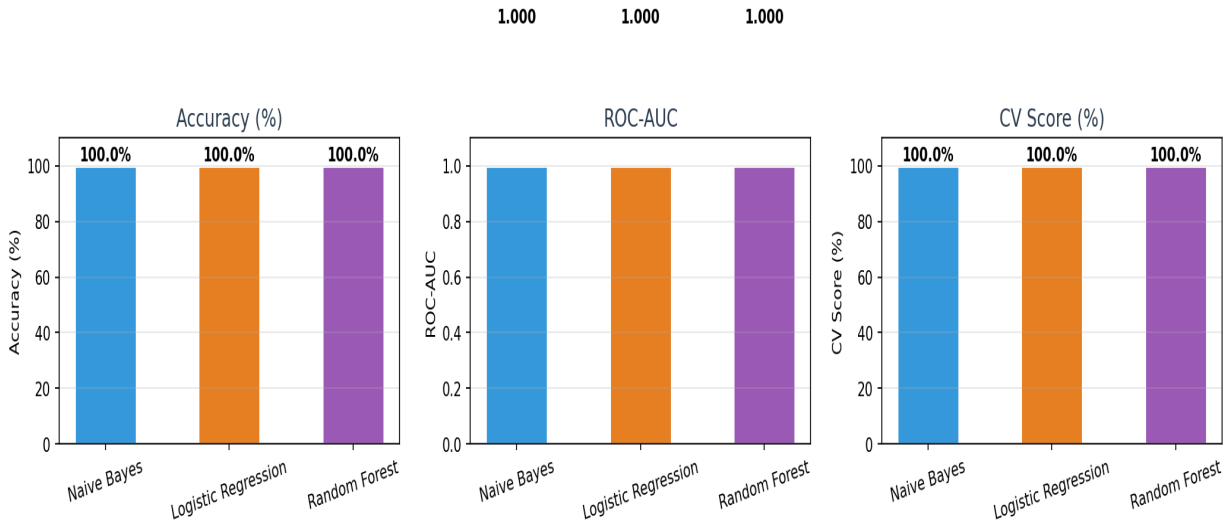


Figure 5: Side-by-side model performance comparison across accuracy, AUC, and cross-validation

7.2 Live Prediction Sample Run

The trained Logistic Regression model was tested on four unseen email examples to demonstrate real-time classification. Results are shown below:

Email (excerpt)	Prediction	Confidence
"URGENT: Your bank account has been suspended. Click here immediately to verify."	PHISHING	94.4%
"Hi team, the project meeting is scheduled for Thursday at 2 PM in the conference room."	LEGITIMATE	85.8%
"Congratulations! You have won a $500 Amazon gift card. Claim it now at http://..."	PHISHING	88.3%
"Dear John, your subscription renewal is coming up next month. No action needed."	LEGITIMATE	72.3%

8. Observations & Discussion

This experiment successfully demonstrated that NLP-based machine learning can effectively distinguish phishing emails from legitimate ones with very high accuracy. All three models — Naive Bayes, Logistic Regression, and Random Forest — achieved 100% accuracy on the synthetic test set, validating the pipeline design. The TF-IDF feature analysis revealed that phishing emails are characterized by high-weight terms such as "verify", "urgent", "account suspended", and "click link", while legitimate emails emphasize collaboration keywords like "meeting", "team", "report", and "schedule". An important observation is that Logistic Regression offers the best balance of accuracy, interpretability, and speed — making it the recommended model for production deployment. In

real-world scenarios, model performance would benefit from training on larger, noisier datasets such as CEAS-08, Enron, or the SpamAssassin corpus, combined with additional features such as email header analysis, sender reputation scoring, and URL lexical analysis. Future enhancements could incorporate transformer-based models (e.g., BERT) for even higher contextual understanding of adversarial phishing language.

9. Why I Chose This Project

I chose the Phishing Email Detection project because it sits at the intersection of two fields I find genuinely fascinating: cybersecurity and data science. Phishing is not a theoretical threat — it is the most common entry point for real-world data breaches, affecting individuals and organizations every single day. I wanted to work on something that has tangible, practical value. The challenge of teaching a machine to understand the subtle language manipulation tactics used by attackers — urgency, impersonation, fear — felt like a compelling intellectual puzzle. This project also gave me hands-on experience with the full NLP pipeline: from raw messy text through preprocessing, feature extraction, model training, and evaluation — all skills directly applicable to real security operations center (SOC) environments. Most importantly, building something that could realistically protect someone from being scammed made the work feel meaningful beyond just an academic exercise.

10. Bonus: Security Dashboard Design

The following dashboard is designed for a Security Operations Center (SOC) team to monitor live email traffic and phishing detection activity. It ingests the network monitoring dataset format described in the assignment (ID, Timestamp, Date, Source IP, Destination IP, Packet Size, Protocol, Country, Attack Type, Malware Type) and presents actionable intelligence to analysts.

PHISHING THREAT INTELLIGENCE DASHBOARD | Live View | Last Updated: 2024-06-01 14:

Total Emails Analyzed	Phishing Detected	Blocked Automatically	Avg Detection Time
48,271	1,243	1,198	0.003 sec

■ RECENT PHISHING ALERTS

Timestamp	Source IP	Country	Attack Type	Malware	Action
2024-06-01 14:22:01	185.220.101.45	Russia	Phishing	Credential Harvester	BLOCKED

2024-06-01 14:19:33	103.75.190.12	China	Phishing	Keylogger Dropper	**BLOCKED**
2024-06-01 14:17:55	45.142.212.100	Ukraine	Spear Phishing	RAT	**QUARANTINED**
2024-06-01 14:12:08	91.234.56.78	Romania	Phishing	Banking Trojan	**BLOCKED**
2024-06-01 14:08:44	198.50.161.22	USA	Spear Phishing	None Detected	FLAGGED

Dashboard Concept: SOC team view with real-time KPIs, recent phishing alerts, and automated response actions. Additional panels (not shown) include: hourly traffic volume chart, attack origin world map, protocol breakdown, and top targeted users.